

```
import pandas as pd
import numpy as np
from scipy import stats
```

```
# Load dataset
```

```
df = pd.read_csv("diabetes.csv")
```

```
print("First 5 records:")
```

```
print(df.head())
```

```
print("\nShape of dataset:")
```

```
print(df.shape)
```

```
print("\nColumn names:")
```

```
print(df.columns)
```

```
First 5 records:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
Shape of dataset:
```

```
(768, 9)
```

```
Column names:
```

```
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness',  
       'Insulin',  
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],  
      dtype='str')
```

```
# Divide data based on diabetes outcome
```

```
non_diabetic = df[df["Outcome"] == 0]
```

```
diabetic = df[df["Outcome"] == 1]
```

```
print("Non-diabetic patients:", len(non_diabetic))
print("Diabetic patients:", len(diabetic))
```

```
Non-diabetic patients: 500
Diabetic patients: 268
```

```
# Create age groups for Chi-square test
```

```
df["AgeGroup"] = pd.cut(
    df["Age"],
    bins=[20, 30, 40, 50, 60, 100],
    labels=["21-30", "31-40", "41-50", "51-60", "61+"]
)
```

```
print("\nAge group distribution:")
print(df["AgeGroup"].value_counts().sort_index())
```

```
Age group distribution:
```

```
AgeGroup
```

```
21-30    417
```

```
31-40    157
```

```
41-50    113
```

```
51-60     54
```

```
61+       27
```

```
Name: count, dtype: int64
```

```
# Sample mean
```

```
sample_mean = df["Glucose"].mean()
```

```
# Sample proportion
```

```
sample_proportion = df["Outcome"].mean()
```

```
print("Sample mean of Glucose:", sample_mean)
```

```
print("Sample proportion of diabetic patients:", sample_proportion)
```

```
print("Percentage of diabetic patients:", sample_proportion * 100)
```

```
Sample mean of Glucose: 120.89453125
```

```
Sample proportion of diabetic patients: 0.3489583333333333
```

```
Percentage of diabetic patients: 34.89583333333333
```

```
# Confidence Interval for Mean
```

```
n = len(df)
```

```
mean = df["Glucose"].mean()
```

```
std = df["Glucose"].std()
```

```
confidence = 0.95
```

```
alpha = 1 - confidence
```

```

# t-critical value
t_critical = stats.t.ppf(1 - alpha / 2, df=n - 1)

# Margin of error
margin_error = t_critical * (std / np.sqrt(n))

lower_mean = mean - margin_error
upper_mean = mean + margin_error

print("95% Confidence Interval for Mean Glucose:")
print(lower_mean, "to", upper_mean)

95% Confidence Interval for Mean Glucose:
118.62972245094552 to 123.15934004905448

# Confidence Interval for Proportion

p = df["Outcome"].mean()

# Z-critical value
z = stats.norm.ppf(1 - alpha / 2)

# Margin of error
margin_prop = z * np.sqrt((p * (1 - p)) / n)

lower_prop = p - margin_prop
upper_prop = p + margin_prop

print("95% Confidence Interval for Diabetic Proportion:")
print(lower_prop, "to", upper_prop)

print("\nPercentage CI:")
print(lower_prop * 100, "% to", upper_prop * 100, "%")

95% Confidence Interval for Diabetic Proportion:
0.3152483405027714 to 0.38266832616389523

Percentage CI:
31.52483405027714 % to 38.26683261638952 %

# One-sample t-test

t_stat, p_value = stats.ttest_1samp(
    df["Glucose"],
    popmean=120
)

print("One-Sample t-Test")
print("T-statistic:", t_stat)
print("P-value:", p_value)

```

```
One-Sample t-Test  
T-statistic: 0.7753502397592414  
P-value: 0.4383717099127051
```

```
# Two-sample t-test
```

```
t_stat, p_value = stats.ttest_ind(  
    non_diabetic["Glucose"],  
    diabetic["Glucose"],  
    equal_var=False  
)
```

```
print("Two-Sample t-Test")  
print("T-statistic:", t_stat)  
print("P-value:", p_value)
```

```
Two-Sample t-Test  
T-statistic: -13.751537067396413  
P-value: 2.644161349540232e-36
```